MIAO LU

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Education

Stanford University *Stanford, USA*
Ph.D. Candidate in Operations Research *Sep.2023 – present*
Advisor: Prof. Jose Blanchet

University of Science and Technology of China *Hefei, China*
B.S. in Mathematics & Applied Mathematics *Sep.2018 – Jun.2022*
Ranking: 2/140, with summa cum laude (Guo Moruo Scholarship)

Industrial Experiences

NVIDIA Research, Research Scientist Intern *Santa Clara, USA*
Host: Dr. Yaosheng Fu and Dr. Zhiding Yu *Feb.2026 – present*
Research on LLM long-context training data synthesis.

ByteDance Seed, Research Scientist Intern *San Jose, USA*
Host: Dr. Jiecao Chen *Jun.2025 – Dec.2025*
Developed end-to-end agentic RL algorithms to scale LLM agent training for long horizon tasks beyond a fixed context limit, with techniques of summarization and context-folding driven context management. Achieved significant improvements over RL baseline in deep research and agentic coding tasks. Progress results in publications [C11][C13] (ACL'26, ICML'26).

Ubiquant Investment, Quantitative Research Intern *Shanghai, China*
Interned at AI department of Ubiquant, research on DL and RL for quantitative trading. *Jun.2022 – Sep.2022*

Research Experiences

Stanford University, Graduate Research Assistant *Stanford, USA*
Advisor: Prof. Jose Blanchet *Sep.2023 – present*
Projects: (i) LLM RLHF from limited data and distributional shifts, the first to propose SFT as regularization in preference learning with theoretical grounding [C9]; (ii) theory and algorithm of distributionally robust decision-making [C6][C8][P2].

Toyota Technological Institute at Chicago, Student Visitor *Chicago, USA*
Host: Prof. Tianhao Wang and Prof. Zhiyuan Li *July.2024 – Sep.2024*
Projects: (i) theoretical understanding of heavy-ball momentum acceleration with large learning rates in river-valley loss landscapes [P3]; (ii) optimal computational-statistical trade-off of learning single-index models via neural networks [C10].

Northwestern University, Remote Research Assistant *Remote*
Host: Prof. Zhaoran Wang *Sep.2021 – Aug.2023*
Selected projects: Principled exploration for online RL under function approximations via value-incentivized regularization (Maximize to Explore [C5]). The method inspires a long line of following work in online/offline RL, multi-agent RL, RLHF.

Invited Talks

Scaling Long Horizon LLM Agent via Reinforcement Learning [C11][C13]

- TikTok Research Seminar, remote [P4][P5] *Oct.2025*

Theoretical Foundations of Distributionally Robust Decision Making [C6][C8][P2]

- 2025 INFORMS annual meeting, Atlanta GA, USA [P1][P2] *Oct.2025*
- 2024 INFORMS annual meeting, Seattle, WA, USA [C8] *Oct.2024*
- 58th Annual Conference on Information Sciences and Systems (CISS), Princeton, NJ, USA [C6] *Mar.2024*
- 2023 INFORMS annual meeting, Phoenix, AZ, USA [C6] *Oct.2023*

Computational-statistical Trade-off of Learning Single-index Models via Neural Networks [C10]

- 2nd Mathematics of Modern Machine Learning Workshop (M3L), Vancouver, BC, Canada *Dec.2024*

Publications & Preprints

Authors with * contributed equally to the work, and † represents alphabetical order.

[P5] **INFUSER: Influence-Guided Self-Evolution Improves Reasoning**
Siyu Chen, **Miao Lu**, Beining Wu, Heejune Sheen, Fengzhuo Zhang, Shuangning Li, Zhiyuan Li, Jose Blanchet, Tianhao Wang, Zhuoran Yang
Preprint under conference review

[P4] **A Few Teacher Steps Go a Long Way: Cost-Efficient On-Policy Data Augmentation for Agent Post-Training**
Junze Ye*, Jiayi Cheng*, **Miao Lu***, Michal Mankowski, Jose Blanchet, Mohsen Bayati
ICML Workshop on RL from World Feedback (RlxF) 2026
Preprint under conference review

[P3] **Towards Understanding Momentum Acceleration in River-Valley Loss Landscape**
Miao Lu, Zeyu Bian, Kaiyue Wen, Beining Wu, Siyu Chen, Tianhao Wang, Zhiyuan Li
ICML Workshop on High-dimensional Learning Dynamics (HiLD) 2026 **Spotlight**
Preprint under conference review

[P2] **Robust Assortment Optimization from Observational Data**
Miao Lu, Yuxuan Han, Han Zhong, Zhengyuan Zhou, Jose Blanchet
arXiv preprint, under journal submission

[P1] **Learning an Optimal Assortment Policy under Observational Data**
Yuxuan Han, Han Zhong, **Miao Lu**, Jose Blanchet, Zhengyuan Zhou
arXiv preprint, under review at Management Science (MS)

[C13] **Scaling Long-Horizon LLM Agent via Context Folding**
Weiwei Sun, **Miao Lu**, Zhan Ling, Xuesong Yao, Kang Liu, Yiming Yang, Jiecao Chen
International Conference on Machine Learning (ICML) 2026

[C12] **Towards Theoretical Understanding of Transformer Test-Time Computing: Investigation on In-Context Linear Regression**
Xingwu Chen*, **Miao Lu***, Beining Wu, Difan Zou
NeurIPS Workshop on Foundations of Reasoning in Language Models (FoRLM) 2025
International Conference on Machine Learning (ICML) 2026

[C11] **Beyond the Context Window: Scaling Agentic RL via End-to-end Optimized Context Compression**
Miao Lu, Weiwei Sun, Weihua Du, Zhan Ling, Xuesong Yao, Kang Liu, Jiecao Chen
Association for Computational Linguistics (ACL) 2026

[C10] **Can Neural Networks Achieve Optimal Computational-statistical Tradeoff? An Analysis on Single-Index Model**
Siyu Chen*, Beining Wu*, **Miao Lu**, Zhuoran Yang, Tianhao Wang
International Conference on Learning Representations (ICLR) 2025 **Oral presentation**
NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L) 2024 **Oral presentation**

[C9] **Provably Mitigating Overoptimization in RLHF: Your SFT Loss is Implicitly an Adversarial Regularizer**
Zhihan Liu*, **Miao Lu***, Shenao Zhang, Boyi Liu, Hongyi Guo, Yingxiang Yang, Jose Blanchet, Zhaoran Wang
Neural Information Processing Systems (NeurIPS) 2024
ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET) 2024

[C8] **Distributionally Robust Reinforcement Learning with Interactive Data Collection: Fundamental Hardness and Near-Optimal Algorithm**
Miao Lu*, Han Zhong*, Tong Zhang, Jose Blanchet
Neural Information Processing Systems (NeurIPS) 2024
ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET) 2024

[C7] **Benign Oscillation of Stochastic Gradient Descent with Large Learning Rates**
Miao Lu*, Beining Wu*, Xiaodong Yang, Difan Zou
International Conference on Learning Representations (ICLR) 2024
NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L) 2023

[C6] **Double Pessimism is Provably Efficient for Distributionally Robust Offline Reinforcement Learning: Generic Algorithm and Robust Partial Coverage**
Jose Blanchet†, **Miao Lu**†, Tong Zhang†, Han Zhong†
Neural Information Processing Systems (NeurIPS) 2023
Extended version under major revision at Mathematics of Operations Research (MOR)

[C5] **Maximize to Explore: One Objective Function Fusing Estimation, Planning, and Exploration**
Zhihan Liu*, **Miao Lu***, Wei Xiong*, Han Zhong, Hao Hu, Shenao Zhang, Sirui Zheng, Zhuoran Yang, Zhaoran Wang
Neural Information Processing Systems (NeurIPS) 2023 **Spotlight**
Extended version under review at Operations Research (OR)

[C4] **Pessimism in the Face of Confounders: Provably Efficient Offline Reinforcement Learning in Partially Observable Markov Decision Processes**
Miao Lu, Yifei Min, Zhaoran Wang, Zhuoran Yang
International Conference on Learning Representations (ICLR) 2023

[C3] **Welfare Maximization in Competitive Equilibrium: Reinforcement Learning for Markov Exchange Economy**
Zhihan Liu*, **Miao Lu***, Zhaoran Wang, Michael I. Jordan, Zhuoran Yang
International Conference on Machine Learning (ICML) 2022

[C2] **Learning Pruning-Friendly Networks via Frank-Wolfe: One-Shot, Any-Sparsity, and No Retraining**
Miao Lu*, Xiaolong Luo*, Tianlong Chen, Wuyang Chen, Dong Liu, Zhangyang Wang
International Conference on Learning Representations (ICLR) 2022 **Spotlight**

[C1] **Learning Robust Policy against Disturbance in Transition Dynamics via State-Conservative Policy Optimization**
Yufei Kuang, **Miao Lu**, Jie Wang, Qi Zhou, Bin Li, Houqiang Li
Association for Advancement of Artificial Intelligence (AAAI) 2022

Awards and Honors

Xinhe Scholarship (outstanding undergraduate researchers, School of the Gifted Young, USTC)	*Mar.2023*
Yuanqing Yang Scholarship (top scholarship, School of Mathematical Sciences, USTC)	*Jan.2022*
The 41st Guo Moruo Scholarship (highest honor, USTC)	*Dec.2021*
Chinese National Scholarship (top scholarship, Ministry of Education of China)	*Nov.2019, 2020*

Academic Services

Journal Reviewer: Annals of Applied Probability (AOAP), Operations Research (OR), Management Science (MS), Mathematics of Operations Research (MOR), Transactions on Machine Learning Research (TMLR)

Conference Reviewer: Neural Information Processing Systems (NeurIPS; 2023-2026), International Conference on Machine Learning (ICML; 2024-2026), International Conference on Learning Representations (ICLR; 2024-2026), International Conference on Artificial Intelligence and Statistics (AISTATS; 2025), ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET; 2024), ICML Workshop on Exploration in AI Today (EXAIT; 2025), ICML Workshop on Methods and Opportunities at Small Scale (MOSS; 2025), NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L; 2024), Association for the Advancement of Artificial Intelligence (AAAI; 2025)